Prajwal Khanal

Enschede, The Netherlands · ktm.prajwalkhanal@gmail.com
github.com/prajzwal08 · Google Scholar · prajzwal08.github.io

RESEARCH INTERESTS

I model soil–vegetation–atmosphere interactions, with a focus on water and carbon fluxes. My work spans both ends of the modeling spectrum: process-based land surface models and data-driven deep learning. I am particularly interested in where the two can inform each other.

EDUCATION

Ph.D., Geo-Information Science and Earth Observation (ITC) Feb 2023 – present
University of Twente, The Netherlands

M.Sc., Environmental Engineering Nov 2020 – Dec 2022
Technical University of Munich, Germany. Grade 1.4/5.0 (German scale; 1.0 = best)

B.E., Civil Engineering Nov 2014 – Aug 2018
Institute of Engineering, Tribhuvan University, Nepal. Rank 20/192 (77.7%)

PUBLICATIONS

Peer-reviewed

1. **Khanal, P.**, et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning. *Biogeosciences*, 21, 1533–1548. doi:10.5194/bg-21-1533-2024
2. **Khanal, P.**, et al. (2020). Flood inundation mapping of the Babai River for different return periods using HEC-RAS and GIS. *Journal of the Institute of Engineering*, 15(2). doi:10.3126/jie.v15i2.27639

Under review

3. **Khanal, P.**, et al. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model. *Submitted.*
4. **Khanal, P.**, et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings. *In review.*

Conference contributions

5. **Khanal, P.**, et al. (2026). Bridging the scale gap: daily 10-meter soil moisture, evapotranspiration, and carbon fluxes via multimodal deep learning. *Workshop on Machine Learning for Land and Hydrology*, ECMWF, Reading, UK.
6. **Khanal, P.**, et al. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning. *EGU General Assembly*, Vienna, Austria, EGU23-7670. doi:10.5194/egusphere-egu23-7670
7. **Khanal, P.** (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. *International Conference on Health Engineering in Disaster*.

WORK EXPERIENCE

Research Intern, Max Planck Institute for Biogeochemistry, Jena, Germany Nov 2020 – Feb 2022
- Quantified the relative control of near-surface soil moisture vs. terrestrial water storage on global vegetation functioning using satellite observation only. (published in *Biogeosciences*).

Student Research Assistant, Chair of Hydrology and River Basin Management, TUM Nov 2021 – Mar 2022
- Produced flood inundation maps for the Ouémé River, Benin (FURIFLOOD) and the Naryn River, Kyrgyzstan (ÖkoFlussPlan) using HEC-RAS 1D unsteady-flow models.

Working Student, Jena-Geos Ingenieurbüro GmbH, Munich, Germany May 2021 – Dec 2021
- Built an Excel/VBA tool for greenhouse-gas accounting across stationary combustion, refrigeration, and transport.

Civil Engineer (Freelance), Executive Consulting Engineering and Planner (ECEP), Kathmandu, Nepal 2018 – 2022
- Prepared business plans for community-based water supply projects in Nepal, for the *Department of Water Supply and Sanitation, Government of Nepal*.
- Developed a smart water supply master plan for the development of Lumbini as a smart city, for the *Department of Urban Development and Building Construction, Ministry of Urban Development, Nepal*.

TEACHING EXPERIENCE

Teaching Assistant, Technical University of Munich 2021 – 2022
- Hydrological and Environmental River Basin Modeling; Integrated Water Resources Management; Rapidly Varying Flow (Hydraulics).

Assistant Lecturer, Cosmos College of Management and Technology, Lalitpur, Nepal Nov 2018 – Jan 2021
- Designed and delivered an elective course on Python and GIS for water resources (final-year students); taught tutorials and labs for Fluid Mechanics and Hydraulics.

AWARDS & SCHOLARSHIPS

- **Second place**, Hackathon on Machine Learning for Earth System Modeling, Aug 2026. Project: *Quantization in foundation models*. Supported by CESOC, ECMWF, TRA Modelling (Univ. Bonn), Forschungszentrum Jülich and Univ. Cologne.
- **Deutschland stipendium**, national merit scholarship for M.Sc. studies, TU Munich, 2021/22.
- **IOE Merit-Based Scholarship**, merit scholarship for B.E. studies, Tribhuvan University, 2014–2018.

TECHNICAL SKILLS & LANGUAGES

Programming	Python, R, MATLAB, VBA, Bash
ML & data	PyTorch, scikit-learn
Geospatial	Google Earth Engine, GIS, GDAL, HEC-RAS
Tools	Git, HPC/SLURM, LaTeX
Languages	Nepali (native), English (C2), Hindi (fluent), Dutch (A2), German (A1)